Predicting Standing Waves on a Square Plate

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Abstract

This investigation aims to use a computer program to predict the Chladni setup for a square metal plate. The plate is vibrated from the middle using function generator. Standing wave patterns on the plate are observed for multiple frequencies. A Python program is written and aims to simulate this setup and predict the patterns. Multiple algorithms are tried and non of them can produce patterns which look identical to the experiment. However, some algorithms predicted certain frequencies really well. Further improvements of the algorithm with better hardwares can be done to obtain more accurate results in the future.

1 Introduction

This research aims to predict the node patterns formed on a vibrating square metal plate. Just like how standing waves can be formed on a string, it can also form on a 2D surface when the plate is vibrating at a certain frequency. This was first demonstrated in public by a German physicist and musician, Ernst Florens Friedrich Chladni.[1] In his original setup, he fixed a metal plate in the centre and used a violin bow to vibrated the edge of the plate. At certain frequencies, standing waves will be formed on the plate. Chladni visulaised this by sprinkling sand on to the plate. The antinodes oscillates which displaces the stand to the nodes, which are still. The sand accumulates in the nodes which forms impressive patterns on the plate. This is later known as the "Chladni Setup".

Chladni Setup is recreated in this experiment by fixing the centre of the plate to an oscillator which is controlled by a function generator. The frequency and the amplitude of the vibration can be controlled directly using the function generator.

This investigation tries to predict the patterns formed on a square plate by using different algorithms explained in the next section.

2 Theory

2.1 Analytical Solutions of the 2D Wave Equation

The first algorithm treats the metal plate as a thin membrane. It has negligible thickness and no elasticity. This way, the waves on the plate simply satisfies the 2D wave equation:[2]

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u \tag{1}$$

The symbol u is a function of t, time, and x,y which are Cartesian coordinates. It represents the vertical displacement of the wave at the given position and time. The ∇^2 symbol is the 2D laplacian operator. The quantity c is the wave speed.

When the wave is a standing wave on the plate, we can assume that the time and space sections of u is separable, $u(x, y, t) = F(x, y)T(t)$, with the time dependent part $T(t) = cos(\omega t)$. Also for convenience, x,y, and t will be casted in dimensionless form. Time t will be multiplied by $\frac{L}{c}$. x and y will all be divided by L, and u represents a fraction of its amplitude. All of the quantities in the paper will be expressed in this dimensionless form later on.

In this way, the wave equation can be rewritten into the time independent form:

$$\nabla^2 u = -k^2 u \tag{2}$$

This is essentially a differential equation and one is trying to find the eigenfunctions of the laplacian operator. One can obtain ω by multiplying k with $\frac{c}{L}$.

An important detail for this PDE is it's boundary conditions. Normally when solving this type of wave equations for a vibrating string, or the Schrodinger's Wave Equation, a Dirichlet boundary condition is assumed. This means that $u = 0$ when $x = 0$ or $x = 1$. This corresponds to the case where the wave must disappear at the edges, i.e. clamped strings or infinite potential. However, for a vibrating plate, the edges are allowed to vibrate freely, often vibrating at a maximum or minimum. Therefore, instead of $u(0) = 0$, it is $\frac{du}{dx}_{x=0} = 0$ (Also the same for x = 1). This is called as the Neumann boundary condition. Both boundary

[1] Linqi 2018.
[2] Emmanuel 2010.

conditions are tested in the practical for the this investigation

Therefore, using this boundary condition, the general solution of this PDE is this:

$$u(x,y,t) = U_0 cos(\omega_{ms} t) cos(m\pi x) cos(s\pi y) \quad (3)$$

U_0 is the initial displacement (i.e amplitude) of the wave. m and s are the indices for the different modes. ω_{ms} is the angular frequency of the wave and is related to m and s via this relationship: $\omega_{ms} = \sqrt{m^2+s^2}$. Note that a value of ω can correspond to multiple combinations of m and s. This means that in the practical, the overall pattern displayed is going to be a superposition of the different modes.

Another important feature of this solution is that each mode is orthogonal to each other. This is key thing to remember when dealing with the discretised version of the laplacian, especially when considering the boundary conditions.

2.2 Numerical Solution of the Wave Equation

Analytical solutions may be convenient for certain situations, however, not any shape has an analytical solution. Sometimes, numerical methods are required. To do so, one must discretise the laplacian (e.i. the second derivative). Dividing the region by N, one can turn the dx into Δx, with $N\Delta x = 1$. This means that the function $u(x)$ is also discretised into u_i where i is an integer ranges from 0 to N. Only one dimension is considered in this situation as it simplifies things a lot. The method to raise this to 2D will be shown later.

2.2.1 Dirichlet Boundary Condition

Using the central difference approach, the second derivative of a point u_i can be thought as the rate of change between the derivative of the imaginary point $u_{i+0.5}$ and the imaginary point $u_{i-0.5}$.

$$\frac{\Delta^2 u_i}{\Delta x^2} = (\frac{\Delta u_{i+0.5}}{\Delta x} - \frac{\Delta u_{i-0.5}}{\Delta x})\frac{1}{\Delta x} \quad (4)$$

As the derivatives of $u_{i+0.5}$ and $u_{i-0.5}$ can be defined like this,

$$\frac{\Delta u_{i+0.5}}{\Delta x} = \frac{u_{i+1} - u_i}{\Delta x}; \frac{\Delta u_{i-0.5}}{\Delta x} = \frac{u_i - u_{i-1}}{\Delta x} \quad (5)$$

one can rewrite the laplacian operator like this:

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} = -\omega^2 u_i \quad (6)$$

Or in a matrix form:

$$\begin{pmatrix} 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} u_{i+1} \\ u_i \\ u_{i-1} \end{pmatrix} = -\Delta x^2 \omega^2 u_i \quad (7)$$

Because the different u_i can be grouped into one vector like this:

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{pmatrix} \quad (8)$$

the matrix equation can therefore be rewritten again into this form:

$$\begin{pmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & & \vdots \\ 0 & 1 & -2 & 1 & & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & \ddots & 1 \\ 0 & \cdots & \cdots & 0 & 1 & -2 \end{pmatrix} \mathbf{u} = -\Delta x^2 \omega^2 \mathbf{u} \quad (9)$$

Notice that u_0 and u_N are both neglected in this vector as they are all 0 in this boundary condition.

This matrix equation turns the PDE into a simpler eigenvalue and eigenvector problem. Notice that the coefficient $-\Delta x^2 \omega^2$ is simply the eigenvalue and $\mathbf{u}$ is the eigenvector of the laplacian matrix. The different eigenvector solutions in this equation is also orthogonal (where the inner product is the dot product instead of the integral). The meaning is that each eigenvector is perpendicular to each other in their ultra high dimensional space (N-1 dim in this case). This is exactly the same ideas as the orthogonal property existing in the continuous, analytical solutions of the equation.

A computer can easily calculate the eigenvector and eigenvalue of this diagonal matrix given enough time and memory.

2.2.2 Neumann Boundary Condition

The matrix shown previously is the discrete Dirichlet Laplacian. The case for free edges is the discrete Neumann Laplacian. As mentioned before, Neumann boundary condition assumes zero flux, i.e. $\frac{du}{dx} = 0$, for $x = 0$ or $x = 1$. When discretised, a ghost point (non existent but imagined) u_{-1} or u_{N+1} can be used to represent the derivative.

$$\frac{\Delta u_0}{\Delta x} = \frac{u_1 - u_{-1}}{2\Delta x} = 0; \frac{\Delta u_N}{\Delta x} = \frac{u_{N+1} - u_N}{2\Delta x} = 0 \quad (10)$$

This implies that $u_1 = u_{-1}$ and $u_N = u_{N+1}$. By applying this to (6), this means that the second derivative of the boundaries is this:

$$\frac{u_1 - 2u_0 + u_{-1}}{\Delta x^2} = \frac{2u_1 - 2u_0}{\Delta x^2} = -\omega^2 u_0 \quad (11)$$

$$\frac{u_{N+1} - 2u_N + u_{N-1}}{\Delta x^2} = \frac{-2u_N + 2u_{N-1}}{\Delta x^2} = -\omega^2 u_0 \quad (12)$$

The change to the original matrix equation is that the first row is $\begin{pmatrix} -2 & 2 & 0 & \cdots & 0 \end{pmatrix}$ and the last row is $\begin{pmatrix} 0 & \cdots & 0 & 2 & -2 \end{pmatrix}$. The matrix also changes from $N-1$ by $N-1$ to $N+1$ by $N+1$. The full matrix equation is shown bellow:

$$\begin{pmatrix} -2 & 2 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & & \vdots \\ 0 & 1 & -2 & 1 & & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & \ddots & 1 \\ 0 & \cdots & \cdots & 0 & 2 & -2 \end{pmatrix} \mathbf{u} = -\Delta x^2 \omega^2 \mathbf{u} \tag{13}$$

Notice that the vector $\mathbf{u}$ now starts with u_0 and ends with u_N. There is a problem for this matrix though. The Laplacian operator is a Hermitian operator which gives it the feature of orthogonal eigenfunctions. The corresponding matrix must also be a Hermitian matrix. This means that the complex conjugate of the transverse of the matrix must be the same as the original matrix. As the matrix shown above is real, this means that the discrete laplacian must have a reflectional symmetry through its main diagonal. The matrix shown above is clearly asymmetrical through the main diagonal. This is not a huge problem mathematically but does put a lot of computational pressure on the computer when it is trying to solve for the eigenvalues and eigenvectors.

To avoid this problem, forward/backward differences can be used instead of central differences method when calculating the second derivatives for the boundaries. Hence, instead of applying the equation shown in (4), one should use this for the left boundary (forward difference):

$$\nabla^2 u_0 = \left(\frac{\Delta u_1}{\Delta x} - \frac{\Delta u_0}{\Delta x}\right)\frac{1}{\Delta x} \tag{14}$$

and this for the right boundary (backward difference):

$$\nabla^2 u_N = \left(\frac{\Delta u_N}{\Delta x} - \frac{\Delta u_{N-1}}{\Delta x}\right)\frac{1}{\Delta x} \tag{15}$$

Using the definition of the Neumann boundary condition, the $\frac{\Delta u_0}{\Delta x}$ term and the $\frac{\Delta u_N}{\Delta x}$ all equals to 0. When backward difference and forward difference is then applied to the left over first derivative terms in the equations respectively, the two second derivatives can be written as follow:

$$\nabla^2 u_0 = \frac{-u_0 + u_1}{\Delta x^2}; \nabla^2 u_N = \frac{u_{N-1} - u_N}{\Delta x^2} \tag{16}$$

Therefore, the overall discrete Neumann laplacian can be expressed as shown:

$$\begin{pmatrix} -1 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & & \vdots \\ 0 & 1 & -2 & 1 & & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & -2 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & -1 \end{pmatrix} \mathbf{u} = -\Delta x^2 \omega^2 \mathbf{u} \tag{17}$$

In this way, the diagonally symmetrical (Hermitian) nature of the laplacian matrix is preserved which made it much cleaner to calculate for the eigenvectors and eigenvalues.

2.2.3 Raising to 2D

The discretised laplacian matrices mentioned previously are all in 1 dimension. This is done intentionally as it is much easier to work with. However, 1D laplacian might be able to model oscillations on a violin string, it is unable to model the 2D waves on a plate. The laplacian must be transformed to its 2D form to model the Chladni setup correctly.

The expanded form of the 2D laplacian is shown:

$$\nabla^2 u = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \tag{18}$$

By imagining a L by L space and dividing the space into a N by N mesh, each discrete point on the mesh can be represented by this notation $u_{i,j}$ where i and j both ranges from 0 to N. As the space is divided into a square mesh, a quantity Δh can be defined so that $\Delta h = \Delta x = \Delta y$. Therefore, using the equations derived previously, the 2 dimensional second derivative can be expressed as following:

$$\frac{u_{i,j+1} + u_{i+1,j} - 2u_{i,j} + u_{i-1,j} + u_{i,j-1}}{\Delta h^2} = -\omega^2 u_{i,j} \tag{19}$$

Flattening the 2 dimensional array of $u_{i,j}$ into 1 column of vector $\mathbf{u}$, the matrix equation of the Dirichlet boundary condition is this:

$$M\mathbf{u} = -\Delta x^2 \omega^2 \mathbf{u} \tag{20}$$

where the vector $\mathbf{u}$ is

$$\mathbf{u} = \begin{pmatrix} u_{1,1} \\ \vdots \\ u_{1,N-1} \\ u_{2,1} \\ \vdots \\ u_{2,N-1} \\ \vdots \\ u_{N-1,N-1} \end{pmatrix} \tag{21}$$

M is a $(N-1)^2 \times (N-1)^2$ tridiagonal matrix

$$M = \begin{pmatrix} B & I & 0 & 0 & \cdots & 0 \\ I & B & I & 0 & & \vdots \\ 0 & I & B & I & & \vdots \\ 0 & 0 & I & \ddots & \ddots & 0 \\ \vdots & & & \ddots & B & I \\ 0 & \cdots & \cdots & 0 & I & B \end{pmatrix} \tag{22}$$

with B being another tridiagonal matrix of size $(N-1) \times (N-1)$ that looks like this:

$$B = \begin{pmatrix} -4 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -4 & 1 & 0 & & \vdots \\ 0 & 1 & -4 & 1 & & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & -4 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & -4 \end{pmatrix} \tag{23}$$

and I is the identity matrix of the same shape with B

The 2 dimensional discrete Neumann Laplacian has a similar equation but with $\mathbf{u}$ being

$$\mathbf{u} = \begin{pmatrix} u_{0,0} \\ u_{0,1} \\ \vdots \\ u_{0,N} \\ u_{1,0} \\ \vdots \\ u_{1,N} \\ \vdots \\ u_{N,N} \end{pmatrix} \tag{24}$$

M is now a $(N+1)^2 \times (N+1)^2$ tridiagonal matrix that looks like this

$$M = \begin{pmatrix} B_1 & I & 0 & 0 & \cdots & 0 \\ I & B_2 & I & 0 & & \vdots \\ 0 & I & B_2 & I & & \vdots \\ 0 & 0 & I & \ddots & \ddots & 0 \\ \vdots & & & \ddots & B_2 & I \\ 0 & \cdots & \cdots & 0 & I & B_1 \end{pmatrix} \tag{25}$$

with B_1 and B_2 displayed bellow respectively.

$$B_1 = \begin{pmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -3 & 1 & 0 & & \vdots \\ 0 & 1 & -3 & 1 & & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & -3 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & -2 \end{pmatrix} \tag{26}$$

$$B_2 = \begin{pmatrix} -3 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -4 & 1 & 0 & & \vdots \\ 0 & 1 & -4 & 1 & & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & -4 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & -3 \end{pmatrix} \tag{27}$$

These matrices might look horrible at first glance but are actually surprisingly easy to construct. Defining the original 1 dimensional matrix as D. Matrix M is simply the kronecker sum of of 2 Ds, with the respective boundary conditions.

$$M = D \oplus D \tag{28}$$

The symbol $\oplus$ denotes the kronecker sum and can be expressed as this

$$D \oplus D = D \otimes I + I \otimes D \tag{29}$$

where the symbol $\otimes$ is the kronecker product of a matrix. This exact detail and proof of this will not be shown in here. See the Appendix for more details on kronecker products and kronecker sums.

3 Methodology

3.1 Chladni Plates

The physical experiment is conducted by attaching a metal plate to a piston, This piston is the controlled by a function generator.